

Aung Khant Ko

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RESEARCH INTERESTS

Dependable Autonomy for Resource-Constrained Robots — *bounded reactive execution for resilient decentralized coordination.*

Reactive execution substrates for robot control: contributing to and extending Rust-native reactive runtimes — typed device hosts, capability-scoped I/O, and embedded ports — for deterministic, heap-allocation-free per-turn execution on resource-constrained hardware.

Failure-aware and supervisory control: runtime detection of actuator nonlinearities, perception faults, and environment-induced failures, with recovery before they corrupt control.

Resilient decentralized coordination: scaling single-robot robustness into multi-robot and swarm settings coordinating collectives and holding formation under noisy sensing, limited computation, and constrained communication.

PUBLICATIONS

Peer-Reviewed Workshop Papers

- [1] Corey Montella, **Aung Khant Ko**, and Steven McPhillimey, *Mech: A Rust-Native Embeddable Reactive Numerical Language for Heterogeneous Computing in Robotics, IROS 2026 Workshop on Why Rust for Robotics? (R4R)*, Pittsburgh, PA, 2026. Accepted for presentation. [OpenReview]

Preprints and Manuscripts

- [2] **Aung Khant Ko**, *Practical Navigation of a Mecanum-Wheel Robot on High-Friction Carpet: Motor Deadzone Compensation, Rotation Asymmetry, and Multi-Phase Goal Management*, engrXiv preprint, 2026. [DOI] *Accepted as a Late-Breaking Result at IEEE/RSJ IROS 2026; withdrawn prior to publication. Extended manuscript in preparation for IEEE Robotics and Automation Letters (RA-L).*

EDUCATION

Thomas Edison State University, Trenton, New Jersey, USA **June 2026**
Bachelor of Arts (B.A.) in Computer Science
Advisor: Dr. Denyse Lemaire

King Mongkut's Institute of Technology Ladkrabang, Bangkok, Thailand **2023 – 2024**
Visiting Student, Robotics and Artificial Intelligence Engineering
Supervisor: Prof. Pitikhate Sooraksa

West Virginia University, Morgantown, West Virginia, USA **2021**
Cultural Exchange Student, Global UGRAD Program (U.S. Department of State)

RESEARCH EXPERIENCE

Independent Research Collaborator, Mech Robotics Lab **Jun 2026 – Present**
Lehigh University, Department of Computer Science and Engineering *Bethlehem, PA*
Supervisor: Prof. Corey Montella

- Investigating Mech v0.4 as a Rust-native reactive runtime for cyber-physical robot control, focusing on persistent state, capability-scoped device I/O, and transactional execution.
- Implemented native RPLIDAR A2 and odometry input hosts that feed typed sensor data into a resident Mech controller and support persistent robot state across reactive turns.
- Integrated a write-capable motor-command host and STM32 interface so Mech programs can issue body-twist commands while Rust handles fixed kinematics and non-bypassable safety interlocks.
- Demonstrated end-to-end real-sensor obstacle avoidance on a physical mecanum robot; a 300-turn run produced five LiDAR-triggered avoidance episodes with 300/300 commands delivered and no runtime safety rejections.
- Measured 2.65 ms median / 3.98 ms p95 sensor-to-command latency and 1.47 ms median / 2.00 ms p95 control-turn compute time at an approximately 10.9 Hz scan-driven cadence.
- Implemented reject-not-clamp runtime checks for command feasibility, freshness, finiteness, and input validity; characterized Mech's atomic validate-or-abort semantics across persistent state and actuation.

- Built an 84-test verification and hardware bring-up suite using emulated LiDAR, motor/odometry sign checks, and reproducible experiments; hardware testing exposed a LiDAR minimum-range blind-zone failure that motivates explicit perception-validity guards.
- Co-authored the accepted IROS 2026 R4R Workshop paper on Mech's embedding and heterogeneous execution model.

Independent Research — Robust Mecanum Navigation on High-Friction Carpet

Nov 2025 – May 2026

Self-Directed Research Project

Bethlehem, PA

Advisor: Dr. Denyse Lemaire, Rowan University

- Investigated model–hardware mismatch in omnidirectional mobile-robot navigation on loop-pile carpet using a physical ROS 2 Jazzy mecanum platform with LiDAR, depth sensing, wheel odometry, IMU, EKF fusion, and AMCL localization.
- Empirically characterized an abrupt motor deadzone that was 75% wider on carpet than on laminate and developed proportional vector scaling that overcomes static friction while preserving mecanum wheel-speed ratios.
- Quantified direction-dependent rotation asymmetry across 80 mm plastic and 100 mm TPU wheel configurations; the plastic-wheel tests exhibited an approximately $9\times$ CW/CCW angular-velocity gap, while TPU wheels reduced the directional bias but produced bimodal behavior.
- Designed Goal Manager V4, a supervisory finite-state architecture that separates heading correction, Nav2 translation, stall recovery, and constrained doorway transit while preserving an unmodified Nav2 stack.
- Achieved 100% navigation success (5/5 trials) on an approximately 14 m multi-room route with autonomous doorway transit; the Nav2-only baseline failed to make progress under the same carpet conditions.
- Published the work as a sole-author engrXiv preprint, accepted as a Late-Breaking Result at IEEE/RSJ IROS 2026 and withdrawn prior to publication; an extended manuscript is in preparation for *IEEE Robotics and Automation Letters (RA-L)*.

Robotics (AI) Research and Development Intern

Apr 2024 – Sep 2024

DHA Siamwalla Ltd.

Bangkok, Thailand

Supervisor: Opas Siamwalla (CEO)

- Researched autonomous mobile robots, sensor integration, and SLAM for warehouse automation.
- Built a TurtleBot3 ROS 1 prototype for autonomous warehouse navigation with A*-based planning, costmaps, and obstacle-aware path execution.
- Developed an OpenCV stereo-vision pipeline for obstacle detection and depth-based localization.
- Integrated TF, odometry, localization, and sensor pipelines for end-to-end robot autonomy; tested in warehouse-aisle navigation and real-time obstacle-avoidance scenarios.

TEACHING EXPERIENCE

STEM Bridge-Building Facilitator, Fun and Learn Education

2026

Northampton Community College

Bethlehem, PA

- Facilitated a hands-on bridge-building activity for children through the Fun and Learn Education program, introducing structural-engineering concepts and the da Vinci self-supporting bridge design.

Computer Science and Robotics Teacher

Mar 2022 – May 2023

Engineering For Kids Myanmar

Yangon, Myanmar

- Taught classes of 15–25 middle- and high-school students, facilitating daily instructional programs, classwork, and assessments.
- Maintained a positive and structured classroom environment, monitoring student behavior and encouraging active participation.
- Developed and adapted STEAM curriculum materials, aligning lessons with student skill levels and learning needs.
- Tracked student attendance, progress, and comprehension through regular quizzes, tests, and assignments.
- Coached a team of 5 students to win a local hackathon; served on the school committee for curriculum development.

Teaching Assistant

Dec 2020 – Feb 2022

Engineering For Kids Myanmar

Yangon, Myanmar

- Supported lead instructors in delivering lessons and managing classroom activities for groups of 15–25 students.
- Provided one-on-one and small-group support to students who needed additional help with assignments and concepts.

- Prepared instructional materials, organized classroom resources, and maintained a clean and functional learning environment.
- Monitored student engagement and reported progress to lead teachers and parents.

Founder & Facilitator, English Language Club
Community Initiative

2021 – 2022
Dawei, Myanmar

- Founded a community English club for non-native speakers, designing conversation-based sessions to build literacy, fluency, and confidence.
- Led weekly group sessions focused on reading comprehension, speaking, and vocabulary development.
- Created a welcoming environment for learners of varying proficiency levels, adapting instruction to individual needs.

HONORS, AWARDS, AND SCHOLARSHIPS

Presidential Alumni Scholarship, Thomas Edison State University
Merit-based scholarship awarded by TESU in recognition of academic achievement.

2026

UPG Sustainability Leader — United People Global
Completed a competitive and immersive field-based sustainability pilgrimage sustainability leadership fellowship facilitated by United People Global (Geneva, Switzerland) and the Hurricane Island Center for Science and Leadership (Maine, USA) on Hurricane Island.

2024

Academic Merit — KMITL Visiting Student Program
Selected as a visiting student in Robotics and Artificial Intelligence Engineering at King Mongkut’s Institute of Technology Ladkrabang, Bangkok.

2023 – 2024

Global UGRAD Scholarship, U.S. Department of State
Competitive merit-based scholarship (Bureau of Educational and Cultural Affairs) for a one-semester academic exchange at West Virginia University. Participated in WVU Experimental Rocketry.

2021

PROFESSIONAL ACTIVITIES

- **IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS 2026)** — Pittsburgh, Pennsylvania, 2026. Conference volunteer and co-author of the accepted R4R Workshop paper, *“Mech: A Rust-Native Embeddable Reactive Numerical Language for Heterogeneous Computing in Robotics.”*
- **Global UGRAD ASEAN Alumni Connections Seminar** — Kuala Lumpur, Malaysia, July 2024. Selected to participate in a multi-day regional conference administered by World Learning and funded by the U.S. Department of State, bringing together Global UGRAD alumni from the ten ASEAN nations for professional development, leadership, and civic-engagement programming.

COMMUNITY & LEADERSHIP

- **Country Representative, Global UGRAD Program** (West Virginia University, 2022 – 2023) — Delivered cultural presentations to elementary- and middle-school audiences, introducing students to Myanmar’s history and diversity.
- **Organizer, “Clean Your Closet” Campaign** (West Virginia University, 2023) — Led a university-wide clothing-donation drive for communities in need in Myanmar.
- **Aspire Leaders Program** (Harvard University) — Completed a leadership-development program with case studies on education, healthcare, and community impact.

SELECTED PROJECTS

- **Mech-Native Cyber-Physical Robot Control Stack** — End-to-end reactive control from real RPLIDAR A2 sensing through resident Mech computation to STM32-based mecanum actuation, with capability-scoped I/O, persistent state, reject-not-clamp runtime safety checks, and measured 2.65 ms median sensor-to-command latency over a 300-turn hardware experiment.
- **Autonomous Mobile Robot (ROS 2 Mecanum)** — ROS 2 Jazzy mecanum robot on Raspberry Pi 5 with EKF sensor fusion, Nav2 (AMCL/NavFn/DWB), custom supervisory Goal Manager, and LiDAR/depth sensing.
- **Sensory Floor Mat for Pediatric Rehabilitation** — Co-designed a multisensory therapy platform with textured tiles and movement-based games at Good Shepherd Rehabilitation Network.
- **TurtleBot3 Stereo-Vision Obstacle Avoidance** — ROS 1 TurtleBot3 with stereo vision, localization, and A* path planning for warehouse navigation.

- **Robot End-Effector Design with ABB IRB1200** — Custom pen end-effector for precise line-drawing tasks on an industrial robot arm at KMITL Robotics Lab.
- **A* Path Planning for TurtleBot3** — A*-based indoor path planning with obstacle avoidance and RViz visualization (ROS 1, Python).
- **Line-Following Robot (STM32)** — Embedded STM32 robot with sensor-based line tracking, PID control, and real-time motor actuation.

CERTIFICATIONS

Robotics Software Engineer Nanodegree	Udacity, Aug 2025
ROS, kinematics, perception, path planning, controls, deep learning for robotics.	
Modern Robotics Specialization (6 courses)	Northwestern University / Coursera, Jun 2025
Rigid-body motion, forward and inverse kinematics, dynamics, motion planning, control.	
Deep Learning Specialization (5 courses)	DeepLearning.AI, 2020 – 2024
Neural networks, CNNs, sequence models, structuring ML projects.	

SKILLS

Spoken Languages: English (Fluent), Burmese (Native)
Programming Languages: Python, C++, C, Rust, Mech, MATLAB
Robotics & Control: ROS 2, ROS 1, Nav2, AMCL, DWB, SLAM Toolbox, Cartographer, TF, robot_localization, Sensor Fusion, EKF, Mecanum Kinematics, PID Control
Cyber-Physical & Reactive Systems: Reactive/Dataflow Control, Persistent-State Control, Capability-Scoped Device I/O, Runtime Safety Interlocks, Sensor-to-Actuator Pipelines
Embedded & Hardware: Raspberry Pi 5, STM32, ESP32, Arduino, RPLIDAR A1/A2, Intel RealSense D435, IMU, Wheel Encoders, Serial Device Integration
Systems & Tools: Linux (Ubuntu 24.04), Rust/Cargo, Git/GitHub, Docker, CMake, Bash, LaTeX, RViz, Gazebo
AI / ML: Deep Learning, CNNs, TensorFlow, Computer Vision (OpenCV), Object Detection
Design: PCB Design (EasyEDA), 3D Modeling (Autodesk Inventor, Fusion 360, Tinkercad), Circuit Analysis (Logisim)